

Norms on Matrix Spaces, Matrix Series, and the Matrix Exponential

Swapnil Ray

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Outline

- 1 Norms on Matrix Spaces
- 2 Convergence of Matrix Series
- 3 The Matrix Exponential
- 4 Algebraic Properties of the Exponential
- 5 Analytic Properties of the Exponential
- 6 Proof of Jacobi's Formula and the Trace–Determinant Formula

Setting and Notation

- $\mathcal{M}_n(\mathbb{C})$: algebra of $n \times n$ complex matrices; $\mathcal{M}_n(\mathbb{R})$: real analogue.
- We identify $\mathcal{M}_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$ when topological/metric notions are needed.
- Goal: introduce a norm compatible with the algebra structure.

Definition: Matrix Norm

Definition 1.1 (Matrix norm)

A *matrix norm* on $\mathcal{M}_n(\mathbb{C})$ is a function $\|\cdot\| : \mathcal{M}_n(\mathbb{C}) \rightarrow [0, \infty)$ such that for all $A, B \in \mathcal{M}_n(\mathbb{C})$ and $\lambda \in \mathbb{C}$:

- ① $\|A\| = 0 \iff A = 0$;
- ② $\|\lambda A\| = |\lambda| \|A\|$;
- ③ $\|A + B\| \leq \|A\| + \|B\|$;
- ④ $\|AB\| \leq \|A\| \|B\|$ (submultiplicativity).

Examples of Submultiplicative Norms

Example 1.2

The following are submultiplicative norms on $\mathcal{M}_n(\mathbb{C})$:

- **Max column sum norm:**

$$\|A\|_1 = \max_{1 \leq j \leq n} \sum_{i=1}^n |a_{ij}|.$$

- **Max row sum norm:**

$$\|A\|_\infty = \max_{1 \leq i \leq n} \sum_{j=1}^n |a_{ij}|.$$

- **Spectral norm:**

$$\|A\|_2 = \sigma_{\max}(A) = \sqrt{\lambda_{\max}(A^*A)}.$$

Verification of Submultiplicativity for $\|\cdot\|_1$

Let $A = (a_{ij})$, $B = (b_{ij})$, $C = AB = (c_{ij})$. Then

$$|c_{ij}| \leq \sum_{k=1}^n |a_{ik}| |b_{kj}|.$$

Fix j and sum over i :

$$\sum_{i=1}^n |c_{ij}| \leq \sum_{i=1}^n \sum_{k=1}^n |a_{ik}| |b_{kj}| = \sum_{k=1}^n |b_{kj}| \sum_{i=1}^n |a_{ik}| \leq \|A\|_1 \sum_{k=1}^n |b_{kj}| \leq \|A\|_1 \|B\|_1.$$

Taking the maximum over j gives $\|AB\|_1 \leq \|A\|_1 \|B\|_1$.

- The other cases are left as an exercise.

Equivalence and a Useful Lemma

Remark 1.3

All norms on the finite-dimensional space $\mathcal{M}_n(\mathbb{C})$ are equivalent, so the choice of norm does not affect convergence.

Lemma 1.4 (Powers and products)

For any submultiplicative norm and $A, B \in \mathcal{M}_n(\mathbb{C})$:

- (i) $\|A^k\| \leq \|A\|^k$ for all $k \geq 1$;
- (ii) $\|A^k B\| \leq \|A\|^k \|B\|$ and $\|BA^k\| \leq \|B\| \|A\|^k$.

Proof. (i) Induction: $\|A^{k+1}\| = \|A^k A\| \leq \|A^k\| \|A\| \leq \|A\|^{k+1}$. (ii) Immediate from submultiplicativity. □

Normalization of the Identity

Remark 1.5

For a submultiplicative norm,

$$\|I\| = \|I^2\| \leq \|I\|^2 \implies \|I\| \geq 1.$$

If $\|I\| \neq 1$, define

$$\|A\|' = \sup_{\|B\| \leq 1} \|AB\|.$$

Then $\|I\|' = 1$ and $\|\cdot\|'$ is still submultiplicative. Hence we may assume $\|I\| = 1$ without loss of generality.

Convergence of Matrix Series

Definition 2.1

Let $(A_k)_{k \geq 0}$ be a sequence in $\mathcal{M}_n(\mathbb{C})$. The series $\sum_{k=0}^{\infty} A_k$ *converges* if the partial sums $S_N = \sum_{k=0}^N A_k$ converge in $\mathcal{M}_n(\mathbb{C})$ (equivalently in $\mathbb{C}^{n^2}$). It *converges absolutely* if $\sum_{k=0}^{\infty} \|A_k\| < \infty$.

Theorem 2.2 (Absolute convergence $\Rightarrow$ convergence)

If $\sum_{k=0}^{\infty} \|A_k\| < \infty$, then $\sum_{k=0}^{\infty} A_k$ converges in $\mathcal{M}_n(\mathbb{C})$, and

$$\left\| \sum_{k=0}^{\infty} A_k \right\| \leq \sum_{k=0}^{\infty} \|A_k\|.$$

Proof of Theorem 2.2

Since $\mathcal{M}_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$ is complete, it suffices to show (S_N) is Cauchy. For $N > M$,

$$\|S_N - S_M\| = \left\| \sum_{k=M+1}^N A_k \right\| \leq \sum_{k=M+1}^N \|A_k\| \rightarrow 0$$

as $M, N \rightarrow \infty$. Hence (S_N) converges to some $S \in \mathcal{M}_n(\mathbb{C})$.

For the norm inequality, continuity of the norm gives

$$\|S\| = \left\| \lim_{N \rightarrow \infty} S_N \right\| = \lim_{N \rightarrow \infty} \|S_N\| \leq \lim_{N \rightarrow \infty} \sum_{k=0}^N \|A_k\| = \sum_{k=0}^{\infty} \|A_k\|.$$

□

Power Series in a Matrix

Theorem 2.3 (Power series in a matrix)

Let $f(z) = \sum_{k=0}^{\infty} c_k z^k$ be a scalar power series with radius of convergence $R \in (0, \infty)$. Then for every $A \in \mathcal{M}_n(\mathbb{C})$ with $\|A\| < R$, the matrix series

$$f(A) = \sum_{k=0}^{\infty} c_k A^k$$

converges absolutely (and hence converges). Moreover, the convergence is uniform on every closed ball $\{A : \|A\| \leq r\}$ with $r < R$.

Proof. For $\|A\| \leq r < R$, Lemma 1.4 gives

$$\|c_k A^k\| \leq |c_k| \|A\|^k \leq |c_k| r^k.$$

Since $\sum |c_k| r^k < \infty$, the Weierstrass M -test yields absolute and uniform convergence on $\{\|A\| \leq r\}$. Letting $r \uparrow R$ gives absolute convergence on $\{A : \|A\| < R\}$. □

Corollary 2.4

Under the hypotheses of Theorem 2.3, the map $A \mapsto f(A)$ is continuous on $\{\|A\| < R\}$; indeed it is analytic there.

Proof. Uniform convergence of a series of continuous (polynomial) functions on each closed ball $\{\|A\| \leq r\}$, $r < R$, implies continuity on each such ball, hence on the union $\{\|A\| < R\}$. Analyticity follows from the same argument applied to the power series expansion about any point A_0 with $\|A_0\| < R$. $\square$

Matrix Exponential

Definition 3.1

For $A \in \mathcal{M}_n(\mathbb{C})$, define

$$\exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

Theorem 3.2 (Convergence of the exponential)

For every $A \in \mathcal{M}_n(\mathbb{C})$, the series $\exp(A)$ converges absolutely. Moreover, for every $r > 0$ the convergence is uniform on $\{A : \|A\| \leq r\}$, and

$$\|\exp(A)\| \leq e^{\|A\|}.$$

Proof of Convergence

Proof. The scalar series $\sum z^k/k!$ has radius $R = \infty$. Apply Theorem 2.3 with $c_k = 1/k!$. The estimate follows from Theorem 2.2:

$$\|\exp(A)\| \leq \sum_{k=0}^{\infty} \frac{\|A\|^k}{k!} = e^{\|A\|}.$$



Proposition 4.1

If $A, B \in \mathcal{M}_n(\mathbb{C})$ satisfy $AB = BA$, then

$$\exp(A + B) = \exp(A) \exp(B).$$

Commuting Case

Proof. Let $S_N(A) = \sum_{k=0}^N A^k/k!$. Since A, B commute, the binomial theorem gives

$$(A + B)^m = \sum_{k=0}^m \binom{m}{k} A^k B^{m-k}.$$

Then

$$S_N(A)S_N(B) = \sum_{j=0}^N \sum_{k=0}^N \frac{A^j B^k}{j!k!} = \sum_{m=0}^{2N} \sum_{\substack{j+k=m \\ 0 \leq j, k \leq N}} \frac{A^j B^k}{j!k!}.$$

Comparing with $S_N(A + B)$ and taking norms,

$$\|S_N(A)S_N(B) - S_N(A + B)\| \leq \sum_{m=N+1}^{2N} \frac{(\|A\| + \|B\|)^m}{m!} \rightarrow 0.$$

Continuity of matrix multiplication gives the result.



Invertibility of the Exponential

Corollary 4.2

For every $A \in \mathcal{M}_n(\mathbb{C})$, $\exp(A)$ is invertible, with

$$\exp(A)^{-1} = \exp(-A).$$

Proof. Apply Proposition 4.1 with $B = -A$ (which commutes with A):

$$\exp(A) \exp(-A) = \exp(A - A) = \exp(0) = I.$$

Similarly $\exp(-A) \exp(A) = I$. Hence $\exp(-A) = \exp(A)^{-1}$. □

Proposition 4.3

For all $A \in \mathcal{M}_n(\mathbb{C})$:

- (i) $\exp(0) = I$;
- (ii) $\exp((s+t)A) = \exp(sA)\exp(tA)$ for all $s, t \in \mathbb{C}$;
- (iii) $\exp(nA) = \exp(A)^n$ for every $n \in \mathbb{Z}$;
- (iv) $\exp(PAP^{-1}) = P\exp(A)P^{-1}$ for every invertible $P \in GL_n(\mathbb{C})$;
- (v) $\exp(A^*) = \exp(A)^*$;
- (vi) $\exp(\overline{A}) = \overline{\exp(A)}$.

Proof. (i) Only the $k = 0$ term survives. (ii) sA and tA commute; apply Proposition 4.1. (iii) Iterate (ii) for $n \geq 1$; use Corollary 4.2 for $n < 0$. (iv) $(PAP^{-1})^k = PA^kP^{-1}$; pass to the limit using continuity of $B \mapsto PBP^{-1}$. (v) $(A^*)^k = (A^k)^*$; conjugation is continuous. (vi) Same argument with complex conjugation. □

Failure of the Product Rule

Example 4.4

Let

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then $A^2 = B^2 = 0$, so

$$\exp(A) = I + A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad \exp(B) = I + B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Hence

$$\exp(A) \exp(B) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad \exp(B) \exp(A) = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

In particular $\exp(A) \exp(B) \neq \exp(B) \exp(A)$, so $\exp(A + B) = \exp(A) \exp(B)$ fails. Note $AB \neq BA$. □

Trace–Determinant Formula

Proposition 4.5

For every $A \in \mathcal{M}_n(\mathbb{C})$,

$$\det(\exp(A)) = e^{\operatorname{tr}(A)}.$$

Proof. *Step 1 (reduction).* By Schur decomposition, $A = UTU^*$ with U unitary and T upper triangular. Then $\exp(A) = U \exp(T) U^*$, so $\det(\exp(A)) = \det(\exp(T))$ and $\operatorname{tr}(A) = \operatorname{tr}(T)$.

Step 2 (triangular case). If T is upper triangular with diagonal entries $t_{11}, \dots, t_{nn}$, then T^k is upper triangular, and $\exp(T)$ is upper triangular with diagonal entries $e^{t_{11}}, \dots, e^{t_{nn}}$. Hence

$$\det(\exp(T)) = \prod_{i=1}^n e^{t_{ii}} = e^{\sum_i t_{ii}} = e^{\operatorname{tr}(T)}.$$

□

Hermitian and Skew-Hermitian Matrices

Proposition 4.6

If A is Hermitian ($A^* = A$), then $\exp(A)$ is Hermitian and positive definite. If A is skew-Hermitian ($A^* = -A$), then $\exp(A)$ is unitary.

Proof. *Hermitian case.* $\exp(A)^* = \exp(A^*) = \exp(A)$, so $\exp(A)$ is Hermitian. By the spectral theorem, $A = U \operatorname{diag}(\lambda_1, \dots, \lambda_n) U^*$ with $\lambda_i \in \mathbb{R}$. Then

$$\exp(A) = U \operatorname{diag}(e^{\lambda_1}, \dots, e^{\lambda_n}) U^*,$$

which has positive eigenvalues $e^{\lambda_i} > 0$. Hence $\exp(A)$ is positive definite.

Skew-Hermitian case.

$$\exp(A)^* \exp(A) = \exp(A^*) \exp(A) = \exp(-A) \exp(A) = \exp(0) = I,$$

so $\exp(A)$ is unitary. □

Derivative Along a Line

Proposition 5.1

For every $A \in \mathcal{M}_n(\mathbb{C})$ and every $t \in \mathbb{R}$,

$$\frac{d}{dt} \exp(tA) = A \exp(tA) = \exp(tA)A.$$

Proof. Consider the difference quotient. Since tA and hA commute,

$$\exp((t+h)A) = \exp(tA) \exp(hA),$$

so

$$\frac{\exp((t+h)A) - \exp(tA)}{h} = \exp(tA) \cdot \frac{\exp(hA) - I}{h}.$$

Now

$$\frac{\exp(hA) - I}{h} = A + h \sum_{k=2}^{\infty} \frac{h^{k-2} A^k}{k!} \rightarrow A \quad (h \rightarrow 0),$$

with the series bounded in norm by $\|A\|^2 e^{|h|\|A\|}$. Hence the limit is $\exp(tA)A$.



Corollary 5.2

For every $A \in \mathcal{M}_n(\mathbb{C})$ and $x_0 \in \mathbb{C}^n$, the unique solution of

$$x'(t) = Ax(t), \quad x(0) = x_0$$

is $x(t) = \exp(tA)x_0$.

Proof. By Proposition 5.1,

$$\frac{d}{dt} \exp(tA)x_0 = A \exp(tA)x_0 = Ax(t),$$

and $x(0) = \exp(0)x_0 = x_0$. For uniqueness, differentiate $e^{-tA}x(t)$:

$$\frac{d}{dt} \left(e^{-tA}x(t) \right) = -Ae^{-tA}x(t) + e^{-tA}Ax(t) = 0,$$

so $e^{-tA}x(t)$ is constant, equal to x_0 at $t = 0$. Hence $x(t) = e^{tA}x_0$. □

Local Diffeomorphism

Proposition 5.3

The exponential map $\exp : \mathcal{M}_n(\mathbb{C}) \rightarrow \mathcal{M}_n(\mathbb{C})$ is a local diffeomorphism near 0. More precisely, there exist open neighborhoods U of 0 and V of I such that $\exp : U \rightarrow V$ is a diffeomorphism.

Proof. For $H \in \mathcal{M}_n(\mathbb{C})$,

$$\exp(H) - I - H = \sum_{k=2}^{\infty} \frac{H^k}{k!},$$

and

$$\|\exp(H) - I - H\| \leq \|H\|^2 e^{\|H\|} = o(\|H\|) \quad (\|H\| \rightarrow 0).$$

Therefore $D\exp(0) = \text{id}_{\mathcal{M}_n(\mathbb{C})}$, which is invertible. By the inverse function theorem, $\exp$ is a local diffeomorphism near 0. □

Jacobian of the Exponential

Theorem 5.4

For every $A, B \in \mathcal{M}_n(\mathbb{C})$, the Fréchet derivative of $\exp$ at A in the direction B is

$$D_B \exp(A) = \int_0^1 e^{sA} B e^{(1-s)A} ds.$$

Proof. Let $\gamma(t) = \exp(A + tB)$. Differentiating the series term by term,

$$\gamma'(t) = \sum_{k=1}^{\infty} \frac{1}{k!} \sum_{j=0}^{k-1} (A + tB)^j B (A + tB)^{k-1-j}.$$

Proof of Theorem 5.4

Setting $t = 0$,

$$D_B \exp(A) = \sum_{k=1}^{\infty} \frac{1}{k!} \sum_{j=0}^{k-1} A^j B A^{k-1-j}.$$

On the other hand, using $\int_0^1 s^p (1-s)^q ds = \frac{p!q!}{(p+q+1)!}$,

$$\int_0^1 e^{sA} B e^{(1-s)A} ds = \sum_{p,q \geq 0} \frac{A^p B A^q}{(p+q+1)!} = \sum_{k=1}^{\infty} \frac{1}{k!} \sum_{j=0}^{k-1} A^j B A^{k-1-j}.$$

□

Real Case: Motivation

Over $\mathbb{C}$, the trace–determinant formula follows from Schur decomposition.

Over $\mathbb{R}$, this argument is unavailable:

- A real matrix need not be (orthogonally) triangularizable.
- Its eigenvalues may be complex.
- Its real Schur form is only block upper triangular with 1×1 and 2×2 blocks.

We instead use *Jacobi's formula* to derive a scalar ODE for $t \mapsto \det(e^{tX})$.

Jacobi's Formula

Theorem 6.1 (Jacobi's Formula)

For any differentiable matrix-valued function $A(t)$ taking values in $\mathcal{M}_n(\mathbb{R})$,

$$\frac{d}{dt} \det A(t) = \operatorname{tr}(\operatorname{Adj}(A(t)) A'(t)) .$$

In particular, if $A(t)$ is invertible,

$$\frac{d}{dt} \det A(t) = \det(A(t)) \operatorname{tr}(A(t)^{-1} A'(t)) .$$

Proof. Write the rows of $A(t)$ as $r_1(t), \dots, r_n(t)$. By multilinearity of the determinant,

$$\frac{d}{dt} \det A(t) = \sum_{k=1}^n \det(r_1, \dots, r_{k-1}, r'_k, r_{k+1}, \dots, r_n).$$

Trace–Determinant Formula for Real Matrices

Expanding along the k -th row,

$$\det(r_1, \dots, r_{k-1}, r'_k, r_{k+1}, \dots, r_n) = \sum_{j=1}^n a'_{kj}(t) C_{kj}(t),$$

where C_{kj} is the (k, j) -cofactor. Since $\text{Adj}(A)_{jk} = C_{kj}$,

$$\frac{d}{dt} \det A(t) = \sum_{j,k} \text{Adj}(A)_{jk} a'_{kj}(t) = \text{tr}(\text{Adj}(A)A'(t)).$$

If A is invertible, $\text{Adj}(A) = \det(A)A^{-1}$, giving the second formula. □

Theorem 6.2

For any matrix $X \in \mathcal{M}_n(\mathbb{R})$,

$$\det(e^X) = e^{\text{tr}(X)}.$$

Proof of Trace-Determinant Formula

Consider $f(t) = \det(e^{tX})$ for $t \in \mathbb{R}$. Since e^{tX} is invertible and $\frac{d}{dt}e^{tX} = Xe^{tX} = e^{tX}X$, Jacobi's formula gives

$$f'(t) = \det(e^{tX}) \operatorname{tr}\left(e^{-tX} \frac{d}{dt}e^{tX}\right) = \det(e^{tX}) \operatorname{tr}\left(e^{-tX} X e^{tX}\right).$$

By cyclicity of the trace,

$$\operatorname{tr}(e^{-tX} X e^{tX}) = \operatorname{tr}(X e^{tX} e^{-tX}) = \operatorname{tr}(X).$$

Thus $f'(t) = \operatorname{tr}(X)f(t)$, a first-order linear ODE with solution

$$f(t) = f(0)e^{t \operatorname{tr}(X)}.$$

Since $f(0) = \det(e^0) = \det(I) = 1$, we get $\det(e^{tX}) = e^{t \operatorname{tr}(X)}$. Setting $t = 1$ yields

$$\det(e^X) = e^{\operatorname{tr}(X)}.$$



Summary

- **Norms:** submultiplicative matrix norms satisfy $\|AB\| \leq \|A\| \|B\|$; all norms on $\mathcal{M}_n(\mathbb{C})$ are equivalent.
- **Series:** absolute convergence implies convergence; power series in matrices converge on balls of radius $< R$.
- **Exponential:** $\exp(A) = \sum A^k/k!$ converges absolutely; $\|\exp(A)\| \leq e^{\|A\|}$.
- **Algebra:** $\exp(A+B) = \exp(A)\exp(B)$ when $AB=BA$; $\exp(A)^{-1} = \exp(-A)$; $\det(\exp(A)) = e^{\text{tr}(A)}$.
- **Analysis:** $\frac{d}{dt} \exp(tA) = A \exp(tA)$; solves $x' = Ax$; local diffeomorphism near 0; $D \exp_A(B) = \int_0^1 e^{sA} B e^{(1-s)A} ds$.
- **Real trace-determinant:** Jacobi's formula + scalar ODE gives $\det(e^X) = e^{\text{tr}(X)}$ for real X .

Thank You!

Questions?